

Abstract : Problem & Purpose : The Vending Machine Simulator is designed to provide an automated and user-friendly system for purchasing products without the need for direct assistance. The project addresses the need for a simple and efficient way to select products, check availability, make payments, and receive the selected product. The simulator demonstrates how technology can be used to automate common purchasing processes and reduce manual effort.

System Overview : System Overview : System Overview : The system allows users to view available products, select a product, check its price and quantity, enter payment, and receive the product along with the appropriate balance or change. It also manages product availability and updates the quantity after a successful purchase. The project can be further improved by integrating technologies such as digital payments, QR codes, UPI, sensors for detecting product dispensing, real-time inventory tracking, cloud connectivity, and mobile applications.

Future Enhancements : Future Enhancements : Future Enhancements : In the future, the system can be enhanced with features such as cashless payment, automatic stock updates, low-stock notifications, personalized product recommendations, remote monitoring, and data analysis. These improvements can make vending machines more convenient, intelligent, secure, and suitable for use in places such as colleges, offices, hospitals, malls, and public areas.